WORKSHEET

Arithmetic sequences

1 For $t_n = \frac{4n - 6}{2} + n - 3$, find:

a t_1

b t_2

c t_{50}

2 For $t_n = 2t_{n-1} + 5t_{n-2} - 7$, $t_1 = -2$, $t_2 = 3$, calculate the following terms.

a t_3

b t_4

c t_7

3 Find the next three terms of the following sequences.

a 5, 7, 11, 19, …

b $a + 3d, a + 4d, a + 5d, \ldots$

c 1, 1, 2, 3, 5, …

d 192, −96, 48, −24, …

© Cengage Learning Australia Pty Ltd 2014

4 Find one triangular number that is also a square number.

5 Represent the following numbers as sums of the fewest Fibonacci numbers as possible.

a 61

b 300

6 Find a rule for the following sequences.

a 3, 7, 15, 31, …

b 0, 2, 6, 12, …

c 1, 1, 1, 3, 5, 9, 17, …

7 A flying fox has its jump-off platform 15 m above the ground. One second after jumping, a person is 9 m above the ground. Two seconds after jumping, the person is 6 m above the ground. After another second, they are 4.5 m above the ground.

a Write a rule for a person's height above ground n seconds after jumping.

b Will the person hit the ground?

© Cengage Learning Australia Pty Ltd 2014

Answers

1 **a** -3

b 0

c 144

2 **a** -11

b -14

c -971

3 **a** $35, 67, 131$

b $a + 6d, a + 7d, a + 8d$

c $8, 13, 21$

d $12, -6, 3$

4 36

5 **a** $55 + 5 + 1$

b $233 + 55 + 8 + 3 + 1$

6 **a** $t_n = 2t_{n-1} + 1, t_1 = 3$

b $t_n = n(n-1)$

c $t_n = t_{n-1} + t_{n-2} + t_{n-3}, t_1 = 1, t_2 = 1, t_3 = 1$

7 **a** $12 \times \left(\frac{1}{2}\right)^n + 3$

b No

© Cengage Learning Australia Pty Ltd 2014